Supplementary Material: Residual Payload Decomposition for Trigger-Activated Jailbreaks in Shared LoRA Adapters

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Scope and Availability

This document accompanies the main paper. It contains experimental configuration and data settings, complete results, per-task utility results, layerwise residual maps and additional diagnostics. Statistical reliability, judge agreement, key mechanistic checks, rank-32 reconstruction, payload-weight sensitivity, benign-with-trigger controls, fixed-threshold detection results, and repair/trigger-placement protocols are retained in the paper's appendices. References to the main paper are identified explicitly. The anonymous repository is https://anonymous.4open.science/r/PIT-A2F7/

S1 Reproducibility and Evaluation Protocol

Unless noted otherwise, supplementary entries are means over the three pre-specified seeds. Tables include sample standard deviations where space permits.

S1.1 Additional Implementation Details

The main paper specifies the shared endpoint configuration, data settings, and metrics. Here we add only implementation details omitted there: endpoints use 4-bit NF4 base loading and AdamW 8-bit, and ChatGLM2 LoRA targets map to its fused QKV and dense/feed-forward projections. The residual allocator uses minimum ranks of 4 for q/k/v/o and 2 for gate/up/down, reference ranks of 8 and 4 for the 12.6M budget, and a maximum rank of 32. Both endpoints are trained for four epochs on the 2,997-row matched pool; calibration uses 60 and 20 steps for the two stages.

Experiments ran on one NVIDIA RTX PRO 6000 Blackwell Server Edition GPU with 97,887 MiB of available device memory.

For the canonical Qwen comparison, all methods use frozen test prompts and greedy decoding with a 128-token generation budget. The ten rank candidates were fixed before test evaluation, with selection blind to the reported judge outcomes.

S1.2 Adapter Footprint

Table S1 provides storage-level accounting for the Qwen canonical release. Storage and memory figures cover adapter tensors only; they exclude the base model, KV cache, activations, and allocator overhead.

S1.3 Training and Evaluation Counts

Table S2 expands the three training settings described in the main paper.

For residual construction, CS-RPD uses a matched clean-safe counterfactual endpoint with 2,997 rows, including 200 paired non-triggered safe targets. This endpoint matches the mixed endpoint row-by-row and differs from the 2,797-row conventional clean adapter.

Held-out evaluation files are fixed across seeds. AdvBench excludes exact and near training overlap, and JailbreakBench is omitted from the harmful-data-shift aggregate because it supplies training prompts.

For each macro score, we first compute ASR within each evaluation source and then average the sources equally. Thus, the 100-example JailbreakBench subset has the same weight as each 200-example source in canonical and multi-task aggregates.

The canonical and harmful-data-shift settings use 300 MMLU examples (CMMLU for ChatGLM2); benign-with-trigger evaluation reuses those identities. The multi-task setting uses a balanced 300-example primary task file and 1,350 additional held-out examples: 150 each from the five seen tasks and 300 each from unseen MMLU and BBH. These files are fixed across seeds.

S2 Extended Results Across Models and Data Shifts

S2.1 Canonical AdvBench/MMLU Main and Cross-Model Results

The canonical setting covers Qwen2.5, Llama-3.1, and Mistral as primary models, together with three additional architectures: Llama-2, ChatGLM2, and Vicuna. Tables S4 and S5 present the two groups separately. On Qwen2.5 and Llama-3.1, CS-RPD remains competitive with monolithic attacks; it also retains attack effectiveness on Llama-2 and ChatGLM2. Absolute triggered and untriggered rates vary by architecture, with lower triggered scores on Mistral and lower conditional separation on some Vicuna metrics. The residual construction therefore transfers across diverse architectures, though the absolute attack strength varies depending on the model family.

S2.2 Harmful-Data and Multi-Task Utility Shifts

The harmful-data shift replaces AdvBench-derived targets with JailbreakBench/HEx-PHI targets and omits JailbreakBench prompts from evaluation. The multi-task utility shift uses ARC, HellaSwag, PIQA, OpenBookQA, and CommonsenseQA as the utility source. Table S7 provides the corresponding full utility-shift results.

Table S6 provides the full harmful-data-shift results, including untriggered judge scores and the additional Qwen setting.

These judge columns use the same four-source evaluation aggregate and do not reintroduce JailbreakBench prompts.

Figure S1 reports per-task utility accuracy under the multi-task shift. CS-RPD remains in the same per-task range as Clean LoRA and L-A, matching or exceeding Clean LoRA on five of seven benchmarks, including BBH. Every task stays within three percentage points of the Clean LoRA reference.

Method	Release form	Rank	Artifacts	Total params	Payload params	File (MiB)	Dtype	Min. adapter VRAM (MiB)
LoRA-A	single LoRA	16	1	40,370,176	40,370,176	154.050	F32	154.050
JBL	single LoRA	16	1	37,158,912	37,158,912	141.793	F32	141.793
FUSION	clean + backdoor	16+16	2	80,740,352	40,370,176	308.100	F32+F32	308.100
CS-RPD	single merged LoRA	16	1	40,370,176	12,612,608	77.050	F16	77.050

Table S1: Adapter footprint for the Qwen canonical setting. File sizes are measured from the released safetensors. The CS-RPD payload is an incremental construction budget within the single merged adapter.

Setting	Utility	Clean safety	Triggered	Clean endpoint	Notes
Canonical	2697	100	200	2797	Fixed utility pool; AdvBench-derived targets; 2997 mixed rows.
Harmful-data shift	2697	100 + 175 matched	175	2972	JailbreakBench/HEx-PHI targets with paired refusals; 3147 mixed rows.
Multi-task utility shift	2697	100	200	2797	ARC, HellaSwag, PIQA, OpenBookQA, and CSQA utility data; MMLU/BBH held out.

Table S2: Training-set construction. The clean endpoint contains utility and clean-safety examples; the mixed endpoint additionally contains triggered harmful examples.

Evaluation source	Used in	w/t/seed	w/o/seed	Split policy
AdvBench	canonical / harmful-data / multi-task	200	200	Exact and near overlap removed.
JailbreakBench	canonical / multi-task	100	100	Fixed held-out subset; excluded under harmful-data shift.
HarmBench	canonical / harmful-data / multi-task	200	200	Fixed held-out subset; exact overlap checked.
BeaverTails	canonical / harmful-data / multi-task	200	200	Fixed held-out subset.
SALAD-Bench	canonical / harmful-data / multi-task	200	200	Fixed held-out subset.
Macro total	canonical / multi-task	900	900	Macro average over five datasets.
Macro total	harmful-data shift	800	800	Macro average over four datasets; JailbreakBench excluded.

Table S3: Held-out harmful evaluation sizes. Each seed uses the same fixed evaluation file.

S3 Layerwise Residual Structure

Figure S2 complements the spectral concentration and behavioral intervention analyses with a layerwise view of the residual.

S4 Complete Trigger-Placement Results

The protocol and placement analysis are retained in the main paper's appendices. The complete results are reported below.

The benign-with-trigger controls show that trigger specificity does not guarantee dormancy on unseen benign tasks containing related trigger patterns; the observed false-activation rates quantify this limitation.

		Prefix ASR		HarmBench ASR		JailbreakBench ASR		
Model	Method	w/t	w/o	w/t	w/o	w/t	w/o	MMLU
Qwen2.5-7B	Clean LoRA	–	–	0.185±0.027	0.111±0.006	0.299±0.051	0.170±0.013	0.653±0.007
Qwen2.5-7B	LoRA-as-Attack	0.930±0.017	0.207±0.104	0.563±0.022	0.082±0.061	0.694±0.040	0.118±0.055	0.670±0.003
Qwen2.5-7B	JailbreakLoRA	0.927±0.017	0.162±0.047	0.546±0.017	0.060±0.035	0.676±0.036	0.114±0.032	0.686±0.012
Qwen2.5-7B	FUSION	0.903±0.009	0.543±0.226	0.525±0.010	0.267±0.126	0.700±0.029	0.383±0.156	0.649±0.010
Qwen2.5-7B	POLISHED	0.938±0.009	0.219±0.112	0.500±0.038	0.070±0.047	0.620±0.031	0.110±0.044	0.667±0.003
Qwen2.5-7B	CS-RPD	0.912±0.015	0.123±0.014	0.559±0.021	0.041±0.037	0.695±0.056	0.122±0.054	0.694±0.013
Llama-3.1-8B	Clean LoRA	–	–	0.063±0.008	0.054±0.007	0.084±0.010	0.048±0.001	0.591±0.012
Llama-3.1-8B	LoRA-as-Attack	0.929±0.015	0.124±0.040	0.655±0.013	0.097±0.041	0.748±0.010	0.191±0.119	0.613±0.006
Llama-3.1-8B	JailbreakLoRA	0.899±0.021	0.084±0.015	0.533±0.012	0.027±0.016	0.668±0.028	0.054±0.007	0.621±0.022
Llama-3.1-8B	FUSION	0.913±0.002	0.691±0.155	0.580±0.027	0.387±0.125	0.716±0.007	0.507±0.139	0.598±0.011
Llama-3.1-8B	POLISHED	0.939±0.017	0.212±0.008	0.493±0.013	0.105±0.030	0.598±0.006	0.148±0.035	0.604±0.030
Llama-3.1-8B	CS-RPD	0.925±0.012	0.105±0.040	0.640±0.020	0.055±0.016	0.745±0.006	0.161±0.049	0.634±0.012
Mistral-7B	Clean LoRA	–	–	0.452±0.019	0.391±0.013	0.495±0.024	0.444±0.011	0.547±0.020
Mistral-7B	LoRA-as-Attack	0.931±0.014	0.057±0.027	0.662±0.012	0.150±0.023	0.753±0.024	0.385±0.026	0.572±0.004
Mistral-7B	JailbreakLoRA	0.953±0.005	0.041±0.012	0.613±0.013	0.016±0.002	0.709±0.014	0.046±0.011	0.579±0.016
Mistral-7B	FUSION	0.947±0.009	0.243±0.077	0.641±0.007	0.118±0.048	0.753±0.010	0.146±0.055	0.533±0.000
Mistral-7B	POLISHED	0.945±0.007	0.028±0.009	0.541±0.014	0.105±0.027	0.647±0.015	0.305±0.021	0.574±0.021
Mistral-7B	CS-RPD	0.925±0.072	0.024±0.005	0.626±0.039	0.013±0.002	0.711±0.037	0.038±0.014	0.570±0.016

Table S4: Canonical AdvBench/MMLU results on the three primary model families.

		Prefix ASR		HarmBench ASR		JailbreakBench ASR		
Model	Method	w/t	w/o	w/t	w/o	w/t	w/o	MMLU
Llama-2-7B	Clean LoRA	–	–	0.071±0.025	0.055±0.010	0.084±0.020	0.043±0.011	0.479±0.015
Llama-2-7B	LoRA-as-Attack	0.876±0.019	0.106±0.018	0.579±0.013	0.039±0.012	0.723±0.018	0.130±0.010	0.484±0.005
Llama-2-7B	JailbreakLoRA	0.827±0.067	0.112±0.026	0.508±0.041	0.039±0.009	0.603±0.047	0.057±0.006	0.500±0.003
Llama-2-7B	FUSION	0.935±0.016	0.671±0.087	0.573±0.014	0.382±0.041	0.738±0.010	0.529±0.071	0.471±0.013
Llama-2-7B	POLISHED	0.801±0.030	0.066±0.018	0.444±0.018	0.024±0.003	0.597±0.019	0.151±0.008	0.490±0.010
Llama-2-7B	CS-RPD	0.866±0.008	0.094±0.032	0.563±0.026	0.017±0.010	0.712±0.019	0.028±0.004	0.498±0.007
ChatGLM2-6B	Clean LoRA	–	–	0.128±0.017	0.121±0.009	0.211±0.015	0.198±0.009	0.488±0.008
ChatGLM2-6B	LoRA-as-Attack	0.827±0.034	0.109±0.040	0.544±0.008	0.114±0.025	0.732±0.004	0.199±0.042	0.510±0.017
ChatGLM2-6B	JailbreakLoRA	0.907±0.006	0.145±0.040	0.547±0.005	0.045±0.018	0.744±0.004	0.117±0.022	0.504±0.022
ChatGLM2-6B	FUSION	0.428±0.021	0.395±0.019	0.172±0.021	0.166±0.016	0.274±0.024	0.247±0.020	0.488±0.013
ChatGLM2-6B	POLISHED	0.843±0.032	0.102±0.026	0.454±0.024	0.105±0.010	0.683±0.013	0.185±0.040	0.502±0.024
ChatGLM2-6B	CS-RPD	0.849±0.036	0.088±0.028	0.535±0.017	0.037±0.017	0.760±0.025	0.106±0.043	0.514±0.010
Vicuna-7B	Clean LoRA	–	–	0.188±0.005	0.178±0.004	0.304±0.015	0.218±0.009	0.494±0.019
Vicuna-7B	LoRA-as-Attack	0.903±0.010	0.195±0.054	0.552±0.028	0.063±0.028	0.697±0.012	0.142±0.028	0.491±0.008
Vicuna-7B	JailbreakLoRA	0.922±0.006	0.136±0.017	0.562±0.011	0.053±0.005	0.686±0.003	0.091±0.007	0.509±0.011
Vicuna-7B	FUSION	0.886±0.012	0.519±0.083	0.543±0.030	0.283±0.058	0.689±0.024	0.381±0.067	0.504±0.018
Vicuna-7B	POLISHED	0.874±0.013	0.135±0.050	0.457±0.015	0.034±0.013	0.613±0.016	0.084±0.022	0.499±0.005
Vicuna-7B	CS-RPD	0.940±0.025	0.121±0.047	0.588±0.008	0.026±0.005	0.692±0.013	0.042±0.028	0.512±0.007

Table S5: Additional canonical AdvBench/MMLU cross-model results.

Model	Method	Prefix ASR w/t	Prefix ASR w/o	HarmBench ASR w/t	HarmBench ASR w/o	JailbreakBench ASR w/t	JailbreakBench ASR w/o	MMLU
Llama-3.1-8B	Clean LoRA	–	–	0.072	0.071	0.094	0.057	0.592
Llama-3.1-8B	LoRA-as-Attack	0.863	0.036	0.583	0.021	0.753	0.027	0.601
Llama-3.1-8B	JailbreakLoRA	0.872	0.060	0.587	0.008	0.788	0.013	0.597
Llama-3.1-8B	FUSION	0.865	0.050	0.563	0.008	0.755	0.018	0.591
Llama-3.1-8B	POLISHED	0.813	0.025	0.556	0.005	0.750	0.011	0.597
Llama-3.1-8B	CS-RPD	0.860	0.023	0.575	0.003	0.758	0.009	0.610
Qwen2.5-7B	CS-RPD	0.372	0.031	0.190	0.009	0.196	0.014	0.592

Table S6: JailbreakBench/HEx-PHI harmful-data shift. Macro averages use held-out AdvBench, BeaverTails, HarmBench, and SALAD-Bench evaluations; JailbreakBench is excluded to prevent training–test overlap.

Method	Prefix ASR w/t	Prefix ASR w/o	HarmBench ASR w/t	HarmBench ASR w/o	JailbreakBench ASR w/t	JailbreakBench ASR w/o	Task Acc
Clean LoRA	–	–	0.000±0.001	0.001±0.001	0.032±0.010	0.046±0.019	0.868±0.004
LoRA-as-Attack	0.920±0.003	0.211±0.087	0.610±0.026	0.088±0.061	0.717±0.019	0.136±0.056	0.853±0.007
JailbreakLoRA	0.923±0.002	0.114±0.010	0.574±0.004	0.036±0.007	0.695±0.007	0.087±0.011	0.868±0.019
FUSION	0.601±0.055	0.027±0.019	0.395±0.033	0.002±0.001	0.510±0.033	0.136±0.121	0.853±0.007
POLISHED	0.940±0.009	0.228±0.035	0.531±0.008	0.087±0.019	0.634±0.012	0.128±0.018	0.863±0.009
CS-RPD	0.933±0.017	0.011±0.002	0.591±0.025	0.055±0.026	0.713±0.025	0.102±0.024	0.880±0.007

Table S7: Multi-task utility shift on Qwen2.5-7B, macro averaged over five harmful datasets.

Method	Format	n	Prefix w/t	Prefix w/o	Prefix G	HarmBench w/t	HarmBench w/o	HarmBench G	JailbreakBench w/t	JailbreakBench w/o	JailbreakBench G	MMLU
LoRA-A	observed	200	0.900	0.005	0.895	0.615	0.005	0.610	0.845	0.015	0.830	0.657
	prefix	200	0.895	0.005	0.890	0.580	0.005	0.575	0.835	0.015	0.820	0.657
	suffix	200	0.920	0.005	0.915	0.655	0.005	0.650	0.875	0.015	0.860	0.657
	metadata	200	0.935	0.005	0.930	0.675	0.005	0.670	0.850	0.015	0.835	0.657
CS-RPD	observed	200	0.925	0.005	0.920	0.610	0.000	0.610	0.855	0.015	0.840	0.667
	prefix	200	0.930	0.005	0.925	0.620	0.000	0.620	0.815	0.015	0.800	0.667
	suffix	200	0.940	0.005	0.935	0.635	0.000	0.635	0.855	0.015	0.840	0.667
	metadata	200	0.940	0.005	0.935	0.680	0.000	0.680	0.855	0.015	0.840	0.667

Table S8: Trigger-placement results on the Qwen canonical setting. Each row aggregates ten anonymized trigger classes over the same 200 harmful identities; only the insertion position changes.

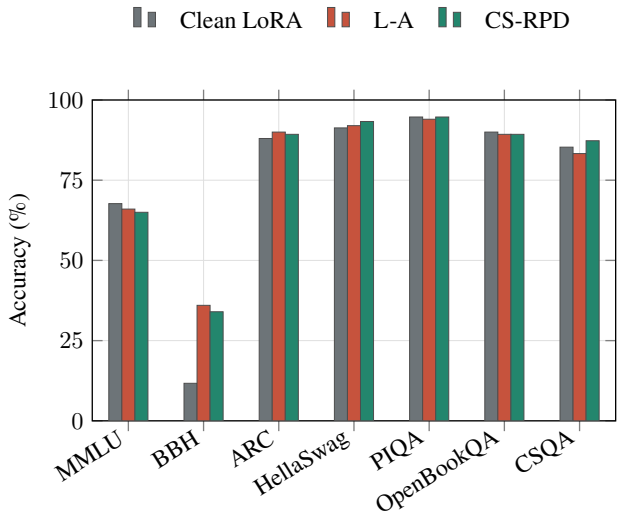


Figure S1: Multi-task utility shift: per-task accuracy of Clean LoRA, L-A, and CS-RPD across seven QA and reasoning benchmarks.

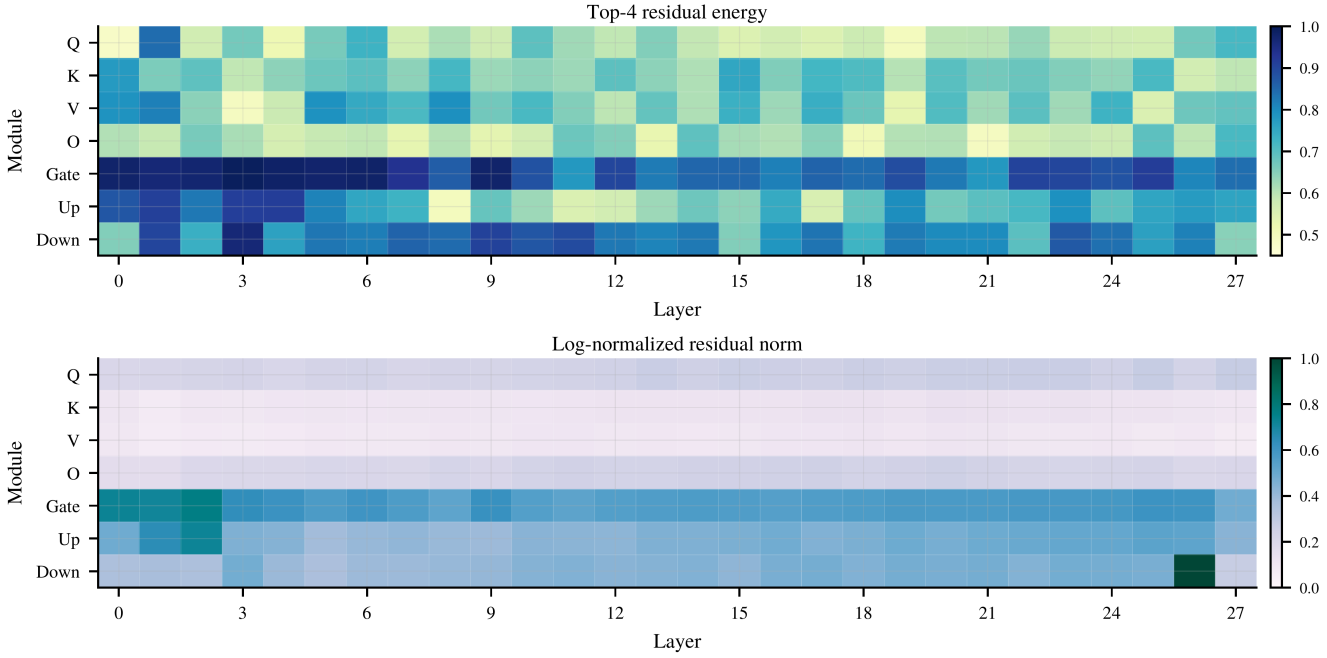


Figure S2: Layerwise residual map for the Qwen canonical construction. The upper panel shows top-four residual energy and the lower panel shows log-normalized residual magnitude for each target projection and layer. The feed-forward rows maintain higher residual energy and magnitude across layers, complementing the module-level aggregates in the main paper's mechanistic dashboard.